

Arduino Microcontroller Experiments

This document contains the Arduino experiments performed using Arduino UNO, Circuit Designer, and the Arduino IDE, with corresponding code and practical evidence.

Experiments included:

1. LED Blinking & GPIO Control
2. IR Object Detection implementation
3. Ultrasonic Distance Measurement
4. OLED Display with Ultrasonic Sensor

Experiment 1: LED Blinking & GPIO Control

Objective:1. Blink an LED 2. Control it using a push button 3. Increase or decrease the LED illumination.

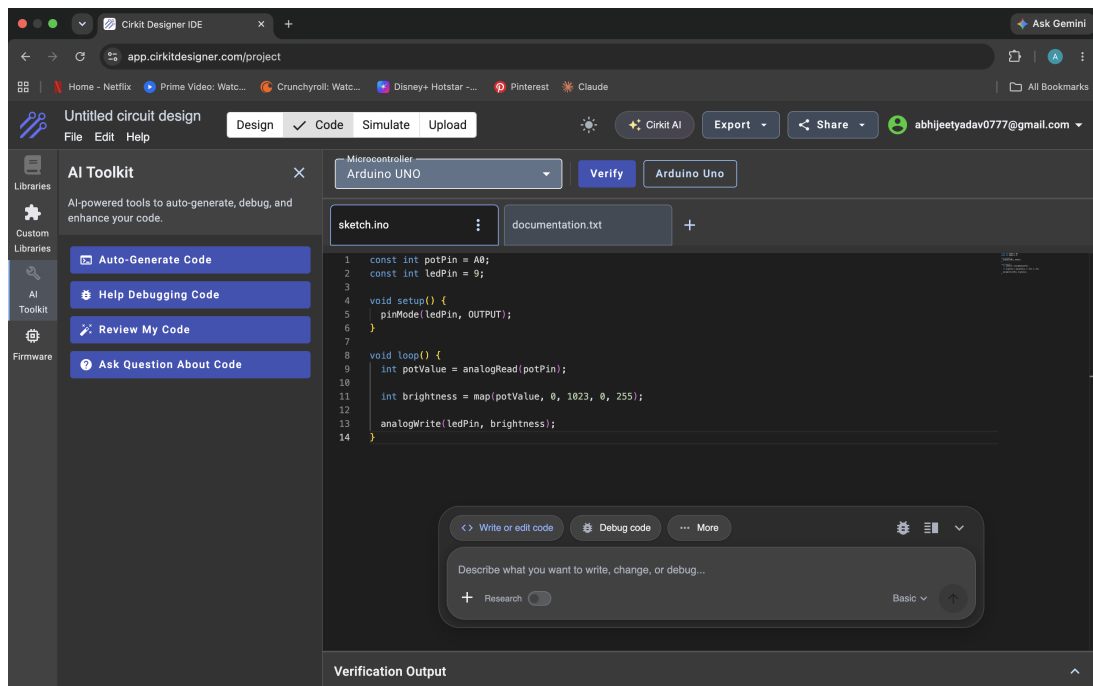
LED illumination control using a potentiometer

```
const int potPin = A0;
const int ledPin = 9;

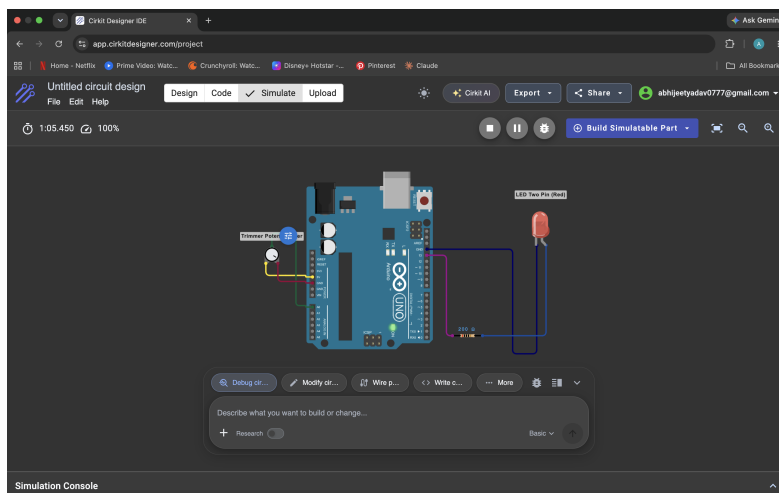
void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  int potValue = analogRead(potPin);
  int brightness = map(potValue, 0, 1023, 0, 255);

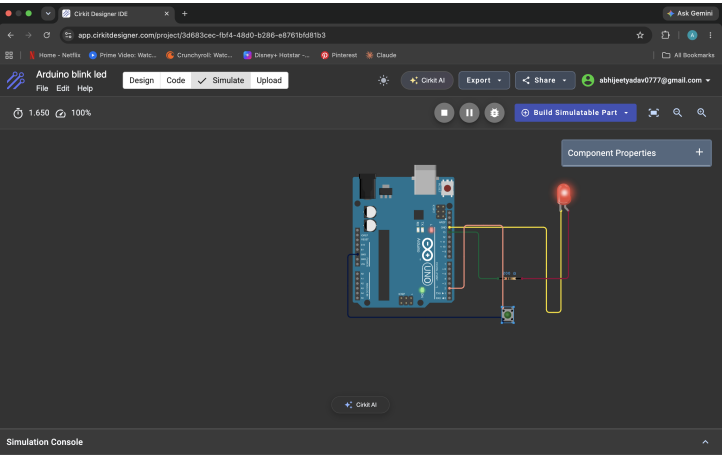
  analogWrite(ledPin, brightness);
}
```



Circuit simulation for LED brightness control:



Push-button control circuit:



Experiment 2: IR Object Detection

Original activity Temperature Monitoring. For the practical implementation, an IR sensor was used instead to detect whether an object was nearby.

Arduino Code

```
const int irPin = 2;
const int ledPin = 13;

void setup() {
  pinMode(irPin, INPUT);
  pinMode(ledPin, OUTPUT);

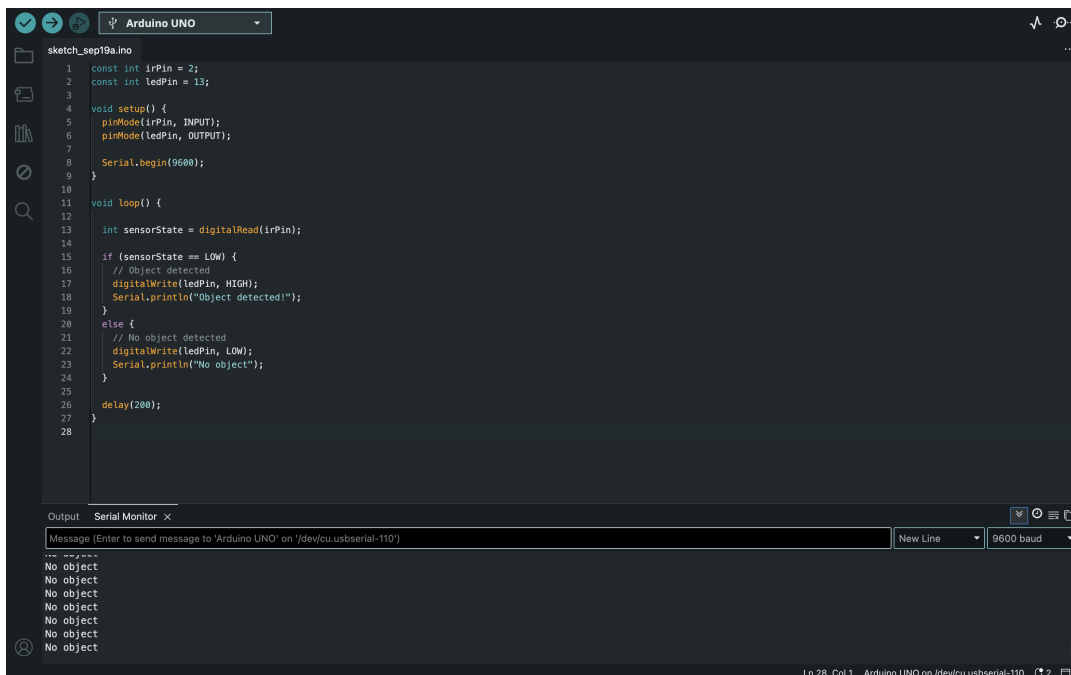
  Serial.begin(9600);
}

void loop() {
  int sensorState = digitalRead(irPin);

  if (sensorState == LOW) {
    // Object detected
    digitalWrite(ledPin, HIGH);
    Serial.println("Object detected!");
  }
  else {
    // No object detected
    digitalWrite(ledPin, LOW);
    Serial.println("No object");
  }

  delay(200);
}
```

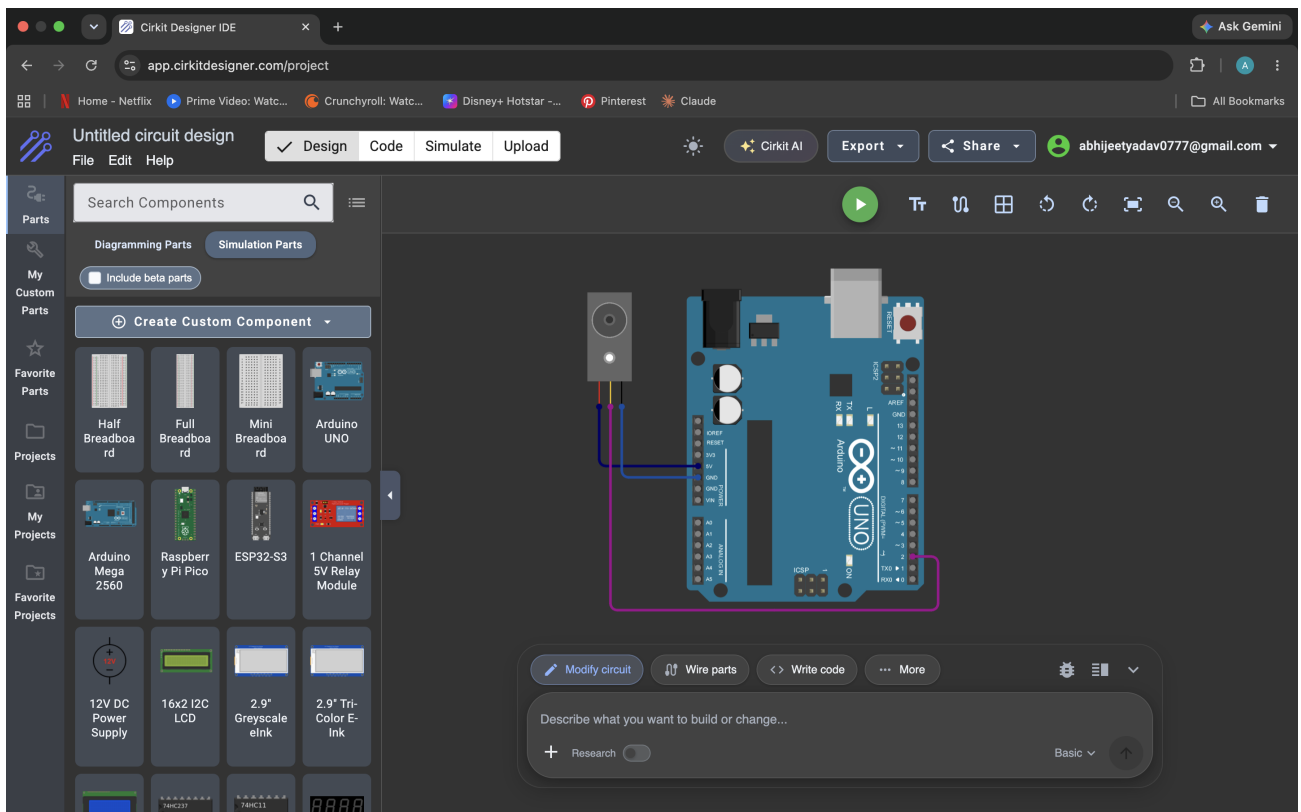
Arduino IDE / Serial Monitor evidence:



Physical IR sensor setup and working output:



Circuit Designer circuit evidence:



Experiment 3: Ultrasonic Distance Measurement

Objective: To measure the distance of an object and display the distance on the Serial Monitor.

Arduino Code

```
const int trigPin = 9;
const int echoPin = 10;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

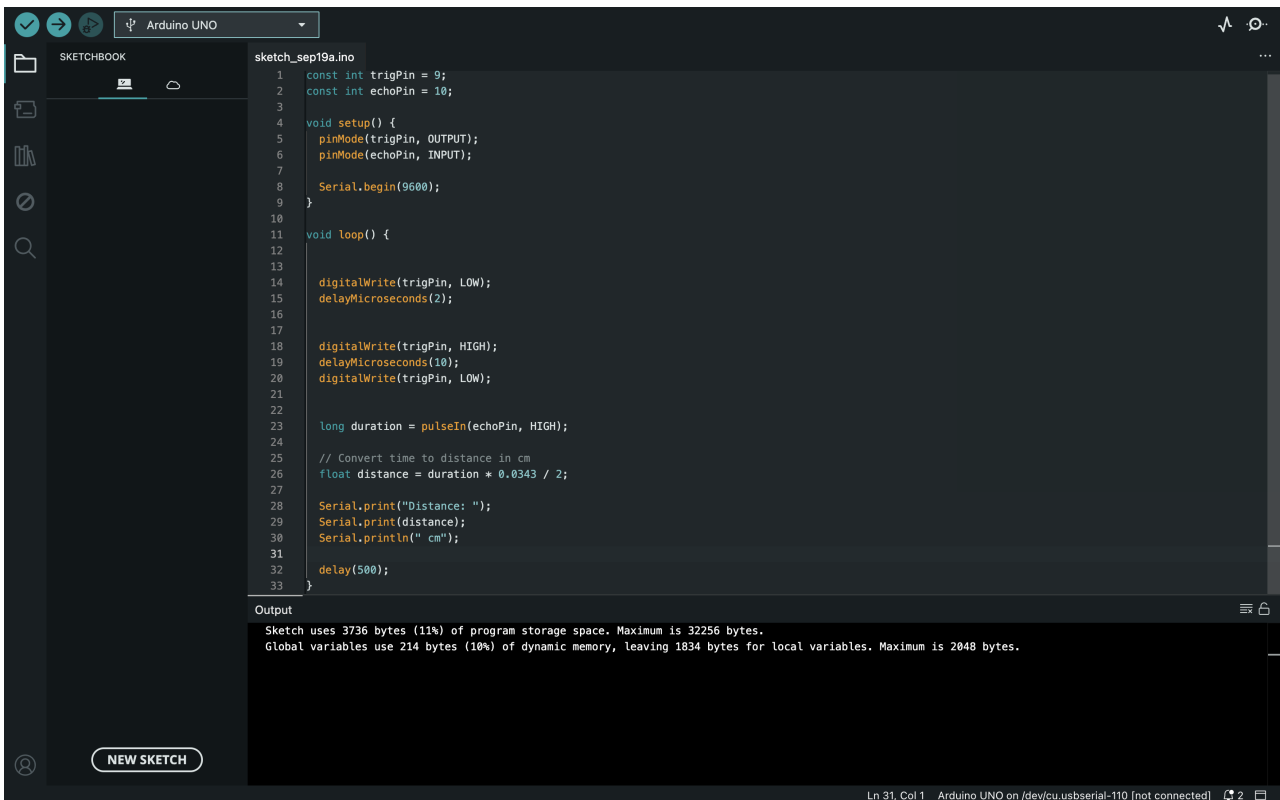
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);

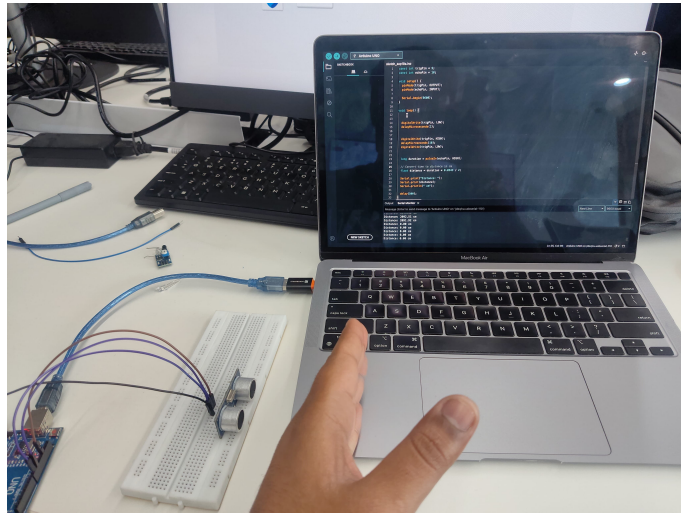
  float distance = duration * 0.0343 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}
```



Physical Arduino UNO + ultrasonic sensor setup:



Experiment 4: OLED Display with Sensor

Objective: Connect a 0.96" OLED, any sensor and display sensor readings and messages.

Arduino Code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

const int trigPin = 9;
const int echoPin = 10;

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  Serial.begin(9600);

  // Start OLED
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found");
    while (true);
  }

  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);

  display.setTextSize(2);
  display.setCursor(0, 0);
  display.println("Ultrasonic");

  display.setTextSize(1);
  display.setCursor(0, 30);
  display.println("Distance Sensor");

  display.display();
  delay(1000);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);

  // Convert to distance in cm
  float distance = duration * 0.0343 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  display.clearDisplay();

  display.setTextSize(1);
  display.setCursor(0, 0);
  display.println("Distance:");

  display.setTextSize(3);
  display.setCursor(0, 20);
  display.print(distance, 1);
  display.println(" cm");

  display.display();

  delay(500);
}
```

Simulation output:

Circuit Designer IDE

app.circuitdesigner.com/project

Untitled circuit design

Design Code Simulate Upload

15.600 100%

Build Simulatable Part

Simulation Console

Distance: 99.38 cm
Distance: 99.50 cm
Distance: 99.38 cm
Distance: 99.38 cm
Distance: 99.38 cm

Message to send here ... Send

The OLED displays the measured ultrasonic distance while the simulation console also prints the distance values.